Cold Storage Unit Equipment Manual

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1. Equipment Identification and Architecture

This manual governs industrial refrigeration units Unit-CR-01 and Unit-CR-02 manufactured for Novacure Labs. Authored by Vikram Shah, Production Manager, with maintenance schedules validated by Arjun Menon.

2. Temperature Control Loop and Tuning

The refrigeration system is engineered to maintain target storage at 2 to 8 °C using redundant dual-stage scroll compressors and variable speed fan coils. Temperature feedback is captured via calibrated Pt100 RTD sensors.

3. Preventive Maintenance Routines

Monthly maintenance includes inspection of evaporator coils, compressor oil levels, door gasket seals, and validation of automatic generator backup switchover circuits.

4. Emergency Response and Failsafe Modes

In the event of compressor malfunction, secondary refrigeration circuits engage automatically within 60 seconds. Alarm sirens and SMS alerts trigger if compartment temperatures drift outside validated boundaries.